INTRODUCTION

Have you heard the sound from FM radio receiver or a beeper or a horn? How the sound is generated? What is the source of the sound? All these questions have only one answer, i. e. the sound is generated by an oscillator.

Any circuit, which is used to generate an ac voltage without an ac input is called an oscillator. The oscillator circuit needs energy from a DC source. It is widely used in electronic equipment. For example, in Radio and Television receivers, oscillators are used to generate high frequency wave called carrier wave. Oscillator generates both sinusoidal (sine) and non-sinusoidal (square, rectangle, triangular, sawtooth, etc.) waveforms.

History of Oscillator


E. H. Armstrong

In 1912, E. H. Armstrong and Lee Deforests developed a new device audion (triode vacuum tube), by coupling one terminal of the device to another. Armstrong achieved the first electronic amplifier with large gain. He called the process "regeneration" (positive feedback). Further, he had made the first electronic oscillator. These two made a revolution in radio broadcasting.

Armstrong had created components necessary to make continuous wave (CW) radio practicable. Primarily, at that time radio was similar to wireless telegraphy (i.e., dots and dashes) and hence CW radio was used to transmit audio information. With Armstrong oscillator, CW signals at high frequencies in the range of kHz to MHz could be easily generated.

7.1. CLASSIFICATION OF OSCILLATORS

Oscillators are classified based on different methods and are summarized in the following sub-sections.

7.1.1. According to the Waveforms Generated

Sinusoidal oscillator: If the output voltage is a sine wave function of time the oscillator is called as "sinusoidal" or Harmonic oscillator. Positive feedback and negative resistance oscillators belong to this category. The waveform generated by the sinusoidal oscillator is shown in Figure 7.1(a)

Non sinusoidal (or) Relaxation Oscillator: This type of oscillators has non-sinusoidal output such as a square, triangular and sawtooth waveforms. The waveform generated by Non sinusoidal oscillator shown in Figure 7.1(b)

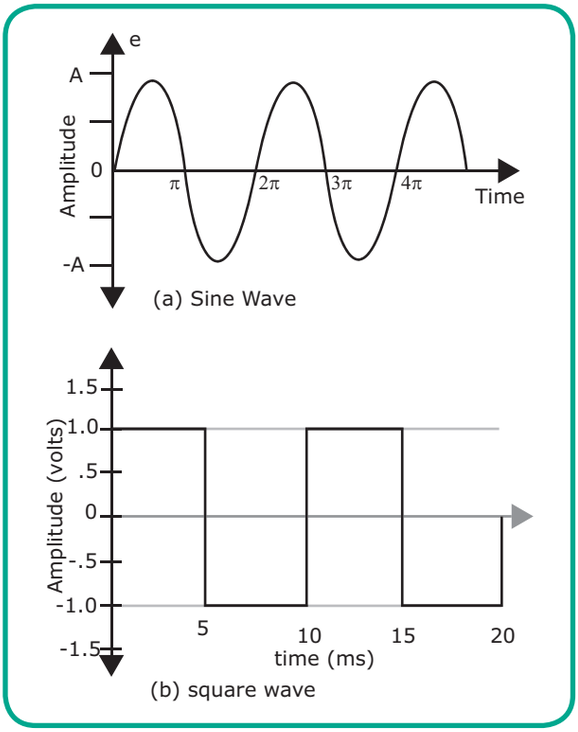


Figure 7.1 (a) Sine Wave, (b) square wave

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